Dual-Camera Active Acquisition for Automated Small-Object Dataset Construction

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Deep learning performance on small objects is frequently bottlenecked by the quality and quantity of training data rather than model architecture. We present a dual-camera active acquisition system that automates small-object dataset generation by pairing a fixed wide-angle camera for persistent detection with a Pan-Tilt-Zoom (PTZ) camera for optical verification. The system runs concurrent YOLO detector instances on an embedded NVIDIA Jetson Orin AGX platform, achieving 35.9 FPS on the GPU for wide-area search and 21.5 FPS on the DLA for PTZ verification. A predictive control framework compensates for mechanical and pipeline latencies to transition from open-loop slewing to closed-loop PID tracking. We demonstrate the system on outdoor drone tracking, where the PTZ autonomously acquires and maintains lock on a small aerial target across a 42-second flight. The active acquisition paradigm replaces digital zoom and super-resolution with true optical zoom, providing optically verified imagery of targets that occupy fewer than 15×15 pixels in the wide field.

I. Introduction

While recent surveys [1, 4] have demonstrated significant architectural improvements in generic object detection, small object detection remains an open problem. The core issue is pixel-level information loss: no amount of digital zoom can recover details that were never sampled. Super-resolution methods attempt to reconstruct these details, but they are computationally intensive and ill-suited for real-time inference on edge hardware [2].

Even with high-resolution imagery, real-time detection pipelines typically downsample inputs to manageable resolutions (e.g., 640×640) to satisfy compute constraints. This reduction inherently compresses distant targets into featureless blobs (Fig. 1) that are difficult to classify without relying on temporal context (as humans do) or additional sensor modalities [3].

Figure 1 illustrates the challenge in distant aerial surveillance, where a target may occupy fewer than 15×15 pixels in a wide-angle feed. At this resolution, distractors (i.e. birds, cloud edges, specular highlights, or sensor noise) are indistinguishable from the target of interest. Standard digital zooming (cropping) only magnifies these ambiguities [5]. To achieve robust detection, a dataset must contain examples where these confusing cases are resolved, yet human labelers often cannot distinguish them in raw wide-angle footage.

A single high-resolution fixed camera does not solve this problem: processing 4K or 8K frames in real time requires prohibitive GPU memory and bandwidth, and distant targets still occupy few pixels relative to the full frame. A single PTZ camera can resolve targets but sacrifices persistent wide-area coverage while zoomed in. The dual-camera configuration decouples these two requirements.

We address the data bottleneck by replacing the human labeler with a dual-camera system that automates the construction of small-object datasets. A fixed wide-angle camera provides persistent coverage, while a controllable Pan-Tilt-Zoom (PTZ) camera provides resolution on demand. The key technical challenge lies in the wide-to-PTZ handover: the system must slew a mechanical lens to capture a moving target based on delayed visual data, compensating for the combined pipeline and mechanical latency.

The contributions of this work are: (i) a hardware-software architecture that synchronizes wide-angle search with narrow-angle verification on an embedded edge platform; (ii) a predictive control formulation that compensates for system latency to enable reliable active target acquisition from wide-angle cues; and (iii) experimental characterization of detector throughput, GPU/DLA parallel inference, and a drone tracking demonstration validating end-to-end system operation.

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Fig. 1 Comparison of Wide-Angle Crop (Digital Zoom), Super-Resolution, and true Optical Zoom active acquisition.

II. Related Work

A. Small Object Detection and Resolution Limits

Classical solutions to the small-object bottleneck involve super-resolution (SR) to reconstruct lost detail from low-resolution inputs [2].

However, reliance on SR for scientific data collection is problematic. First, SR is fundamentally generative; it estimates high-frequency details based on learned priors, creating a risk of hallucination where the model reinforces its own biases [5]. Second, high-fidelity generative models required for scientific validity often require > 300 ms per frame on edge accelerators [18], whereas lightweight models sacrifice reconstruction quality. Our system achieves mechanical slew-and-settle in a comparable time window while obtaining true optical ground truth rather than hallucinated details.

B. Active Acquisition vs. Continuous Tracking

Most PTZ tracking literature focuses on the control problem of keeping a target centered in the frame [7, 8]. This requires mitigating total system latency (video encoding + network + mechanical response), which for IP-based systems frequently ranges from 200–500 ms [17]. Our work addresses a different objective we term *slew-to-classification*: rather than maintaining persistent lock on one target, the system performs discrete acquisitions of ambiguous candidates to verify their identity.

Existing active perception systems like VIGIA-E [11] typically optimize for broad area coverage or anomaly detection. Our system instead functions as a sparse query mechanism: it identifies specific low-confidence candidates in the wide field and commits the PTZ resource to verifying them individually.

C. Sensor-Driven Labeling

Reducing manual annotation is a central goal of both semi-supervised learning and active learning. Pseudo-labeling methods such as ASTOD [12] attempt to retrain models using high-confidence predictions, but this approach often fails in the small-object regime where the detector is consistently uncertain [13]. Active learning strategies like PPAL [15] identify informative samples but still require a human loop [14].

Our active acquisition creates a fully automated hybrid: the selection logic of active learning (targeting less confident samples) is paired with PTZ verification instead of a human annotator. While the target is ambiguous at 15×15 pixels, the zoomed view restores it to a regime ($> 100 \times 100$ pixels) where off-the-shelf detectors achieve 99.1% mAP50 [6]. By physically bridging the gap between the surveillance view and the high-resolution training distribution, we convert a difficult small-object inference problem into a trivial classification task.

Fig. 2 Physical hardware configuration. Two FLIR M300C cameras are mounted on a shared three-foot stand with ≈ 0.5 m baseline separation. The left unit serves as the fixed wide-angle camera; the right unit operates as the PTZ camera.

III. System Overview and Formulation

A. Hardware and Software Roles

The system consists of two FLIR M300C cameras mounted on a shared three-foot stand with a fixed relative pose (Fig. 2): (1) a fixed wide-angle camera providing continuous coverage and running a real-time YOLO-family detector on the primary GPU, and (2) a PTZ camera whose pan, tilt, and zoom are commanded via an HTTP control interface. Both cameras are powered by a 12.8 V 50 Ah LiFePO4 battery, selected because the dual-camera current draw exceeded the capacity of a standard DC power supply unit. The cameras connect to an NVIDIA Jetson Orin AGX (JetPack 4.7, CUDA 12.6) via a PCIe Ethernet network card with IEEE 802.3at PoE detection, confirmed safe for the non-PoE FLIR devices. The baseline separation $b \approx 0.5$ m produces a parallax angle of $\arctan(0.5/100) \approx 0.29°$ at 100 m range, which is negligible for the initial open-loop slew command.

The software architecture is organized in three layers (Fig. 5). The first layer (L1) provides a Kivy-based GUI with a streamer thread displaying dual camera feeds and exposing controls for zoom, movement speed, tracking mode, and PID tuning parameters. The second layer (L2) runs six threads: a capture thread, a YOLO processing thread, and a draw thread for each camera. The third layer (L3) implements the tracker process, which operates in two phases: Phase 1 performs target acquisition from the stationary camera, and Phase 2 transitions to active tracking on the PTZ camera via a coordinate mapper, centering controller, zoom controller, PID controller, and HTTP command interface. If the PTZ loses detection during Phase 2, the system falls back to stationary camera tracking; once the PTZ reacquires the target, it resumes PTZ-based control.

B. Problem Formulation

Let I_t^w be the wide-angle frame captured at time t, and let the PTZ state be $\tau_t = (\theta, \phi, z)_t$. The wide detector produces candidate detections $D_t = \{(b_i^w, s_i^w, c_i)\}_i$. At each decision step, the system selects a target for PTZ verification subject to two constraints: the PTZ must be available (not currently slewing) and the target must lie within the reachable field of view.

The current implementation uses a greedy threshold policy: the system acquires the largest detected target whose confidence exceeds the acquisition threshold $s > \tau_{\text{acq}}$. This policy prioritizes targets that provide the greatest pixel-area gain under optical zoom while filtering out low-confidence false alarms. The action takes effect at time $t + \delta$, where δ accounts for the combined processing and mechanical slew delay, so the PTZ command is issued against a predicted

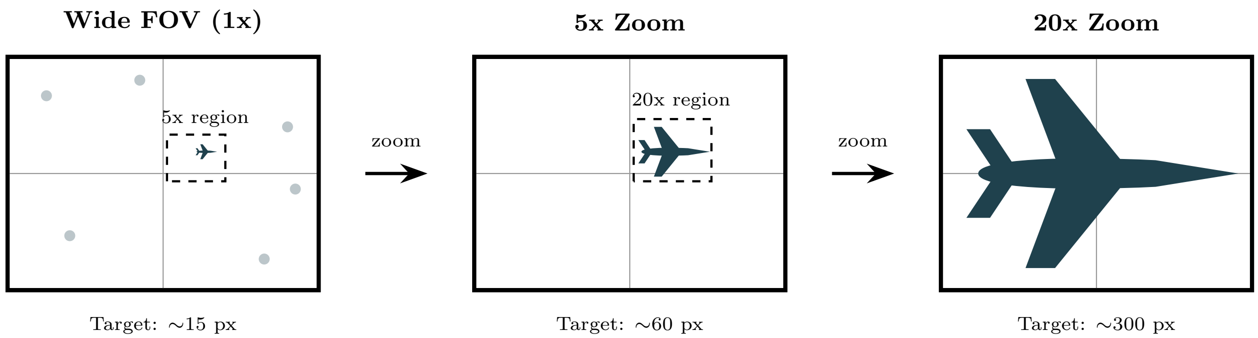

Fig. 3 Visual progression of an active acquisition sequence. A 15-pixel target in the wide field (1x) is indistinguishable from noise. The system cues the PTZ to an intermediate 5x zoom for acquisition, then a 20x zoom for final detailed verification, revealing the aircraft structure clearly.

target position rather than the instantaneous detection center (Section V).

IV. Calibration and Cross-View Geometry

A. Camera Model and Calibration

Both cameras are modeled with intrinsic matrices (K_w, K_p), and R_{pw} denotes the rotation from wide-camera coordinates to the PTZ base frame. Because both cameras are rigidly mounted on a shared stand, the relative orientation is fixed and was measured as a one-time azimuth offset ($-10°$) and elevation offset ($0°$) during installation, then verified against known landmarks.

For sky targets at long range, parallax between cameras is negligible. With a baseline of $b = 0.5$ m and a minimum operating range of $R = 100$ m, the angular parallax is $\arctan(b/R) \approx 0.29°$, well within the PTZ field of view at any zoom level. A direction-only transfer therefore provides a sufficiently accurate initial seed; once the PTZ is slewed to this bearing, its onboard detector refines the tracking lock. Translation is ignored in this regime, and cross-view mapping operates on ray directions on the unit sphere.

B. Mapping Wide Detections to PTZ Pan/Tilt Commands

Let (u, v) be the center of a wide detection box b^w in pixel coordinates. After undistortion, the bearing direction in the wide camera frame is

$$\hat{d}_w = \frac{K_w^{-1}[u\ v\ 1]^\top}{\|K_w^{-1}[u\ v\ 1]^\top\|}. \tag{1}$$

Transforming to the PTZ base frame gives $\hat{d}_p = R_{pw}\hat{d}_w$. Using a conventional yaw–pitch parameterization, the commanded pan (yaw) and tilt (pitch) are

$$\theta = \operatorname{atan2}(\hat{d}_{p,y}, \hat{d}_{p,x}), \quad \phi = \operatorname{atan2}\left(\hat{d}_{p,z}, \sqrt{\hat{d}_{p,x}^2 + \hat{d}_{p,y}^2}\right). \tag{2}$$

Because PTZ motion and video pipelines introduce latency Δ, $\hat{d}_p$ is computed from a short-horizon prediction rather than the instantaneous detection center (Section V).

C. Zoom Selection by Target Angular Size

Let h^w be the detected box height in wide pixels. Under small-angle approximation, the apparent angular height is approximately $\alpha \approx h^w/f_w$, where f_w is the wide focal length in pixels. To obtain a desired PTZ pixel height h^p_{des}, select a PTZ focal length $f_p(z)$ such that

$$h^p_{\text{des}} \approx \alpha f_p(z) \approx \frac{h^w}{f_w} f_p(z), \tag{3}$$

then choose the zoom z whose calibrated $f_p(z)$ best matches the required value, clipped to PTZ limits. This approach avoids explicit range estimation and is well suited for distant aircraft.

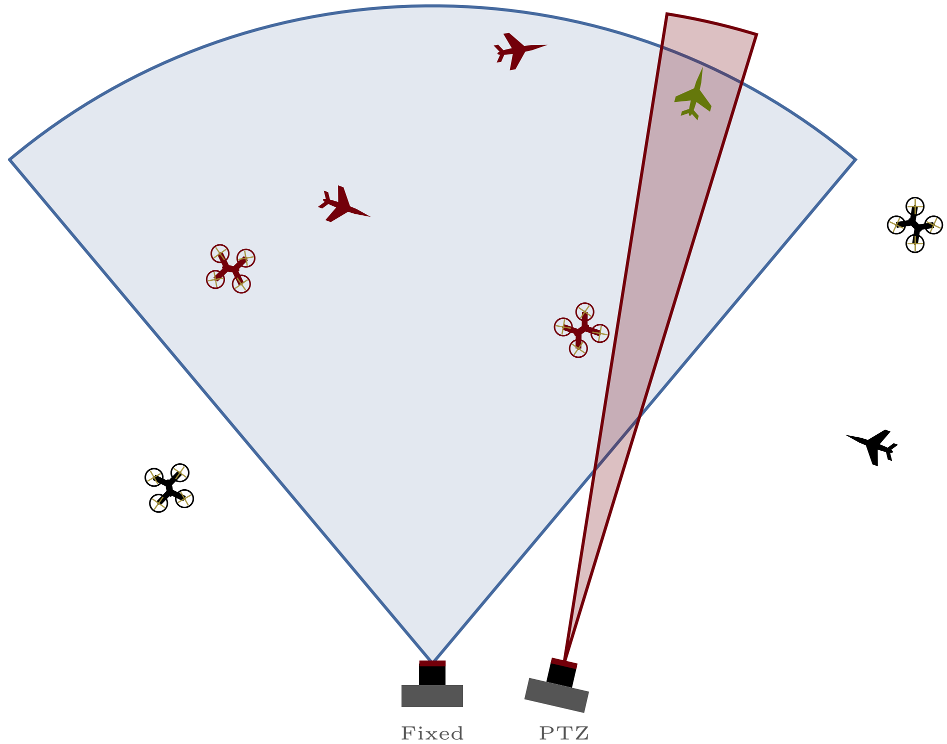

Fig. 4 Field of view comparison. The wide camera (blue) covers approximately 60° horizontally for persistent surveillance, while the PTZ camera (red) provides a narrow, steerable view.

V. PTZ Triggering, Tracking, and Scheduling

We use wide-camera tracklets to provide temporal coherence and to predict target bearing during PTZ motion. The predictor uses a constant-velocity model with exponential moving average (EMA) smoothing. At each detection update, the instantaneous pixel velocity is computed from consecutive detection centers and filtered as $\mathbf{v}_t = \alpha\, \mathbf{v}_{t-1} + (1-\alpha)\, \mathbf{v}_{\text{inst}}$ with smoothing factor $\alpha = 0.3$. The predicted bearing at command-arrival time is then

$$\hat{d}_w(t+\Delta) = d_w(t) + \mathbf{v}_t \cdot \Delta, \tag{4}$$

where Δ is the estimated command latency. This model is sufficient for targets with approximately linear short-horizon trajectories, which is the dominant case for aircraft and drones at range.

A. Triggering Policy

The system triggers PTZ acquisition on the largest detected target whose confidence exceeds the acquisition threshold ($s > 0.40$). Selecting the largest target maximizes the expected pixel-area gain under optical zoom, since larger wide-field detections correspond to closer or more resolvable targets. Because only one PTZ resource is available, the system processes candidates sequentially: if the PTZ is occupied, additional candidates are queued until the current acquisition completes or times out (10 s). This greedy policy is sufficient for the single-target drone tracking scenario evaluated here; multi-target prioritization via uncertainty weighting or novelty scoring is a planned extension.

VI. Automated Dataset Construction

A. Label Sources and Cross-View Transfer

The PTZ view is treated as a higher-fidelity label source when it produces a confirmed detection for the target class. The label transfer module maps the PTZ detection back into the wide image coordinate system.

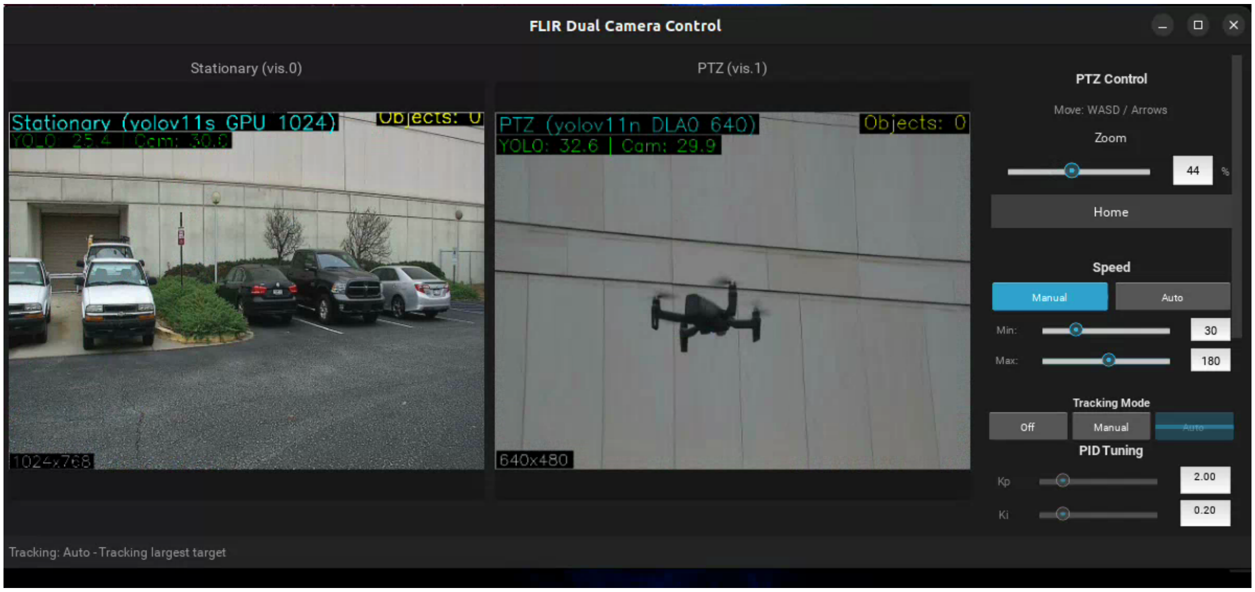

Fig. 5 Live screenshot of the three-layer software interface during a drone tracking test. The stationary camera (left) runs YOLOv11s on the GPU at 1024×1024 resolution for wide-area detection. The PTZ camera (center) runs YOLOv11n on the DLA at 640×640, here tracking the drone at optical zoom. The control panel (right) exposes PTZ zoom, movement speed, tracking mode selection, and PID tuning parameters.

Algorithm 1 Real-time wide-to-PTZ acquisition

Init f_θ (wide), $\mathcal{T}$ (tracker), g (PTZ), $\mathcal{D}$ (data)
for each wide frame I_t^w **do**
  $D_t \leftarrow f_\theta(I_t^w)$; $\mathcal{T} \leftarrow \text{UpdateTracks}(\mathcal{T}, D_t)$
  Filter tracks: $\{o \mid s_o > \tau_{\text{acq}}\}$
  Select $o^\star \leftarrow \arg\max_o \text{area}(b_o^w)$
  if $o^\star$ exists AND PTZ available **then**
    Predict $\hat{d}_w(t+\Delta)$; map to (θ, ϕ, z)
    Command PTZ: $g(\theta, \phi, z)$; get $I_{t'}^p, \tau_{t'}$
    Run PTZ detector: $D_{t'}^p \leftarrow f_{\theta_p}(I_{t'}^p)$
    if Quality OK AND max $s^p > \tau_{\text{confirm}}$ **then**
      $b^{w \leftarrow p} \leftarrow \text{Project}(b^p, \tau_{t'})$
      Commit $(I_t^w, b^{w \leftarrow p}, c)$ to $\mathcal{D}$
    end if
  end if
end for
Periodically retrain f_θ on $\mathcal{D}$

For distant targets, direction-only transfer proceeds by converting the PTZ bounding box corners into bearing rays, rotating those rays into the wide camera frame, and projecting them into the wide image plane. Let (u_k^p, v_k^p) be PTZ box corners; compute PTZ rays $\hat{d}_k^p$ from K_p^{-1}, rotate to wide frame $\hat{d}_k^w = R_{wp}\hat{d}_k^p$, then project:

$$[u_k^w, v_k^w, 1]^\top \propto K_w \hat{d}_k^w, \quad k \in \{1, 2, 3, 4\}. \tag{5}$$

The transferred wide box is the tight axis-aligned rectangle enclosing the projected corners.

B. Quality Gates and Drift Prevention

Self-training and pseudo-labeling can suffer from confirmation bias if low-quality pseudo-labels are admitted. In this system, the primary drift control mechanism is the physical zoom verification itself: ambiguous wide detections

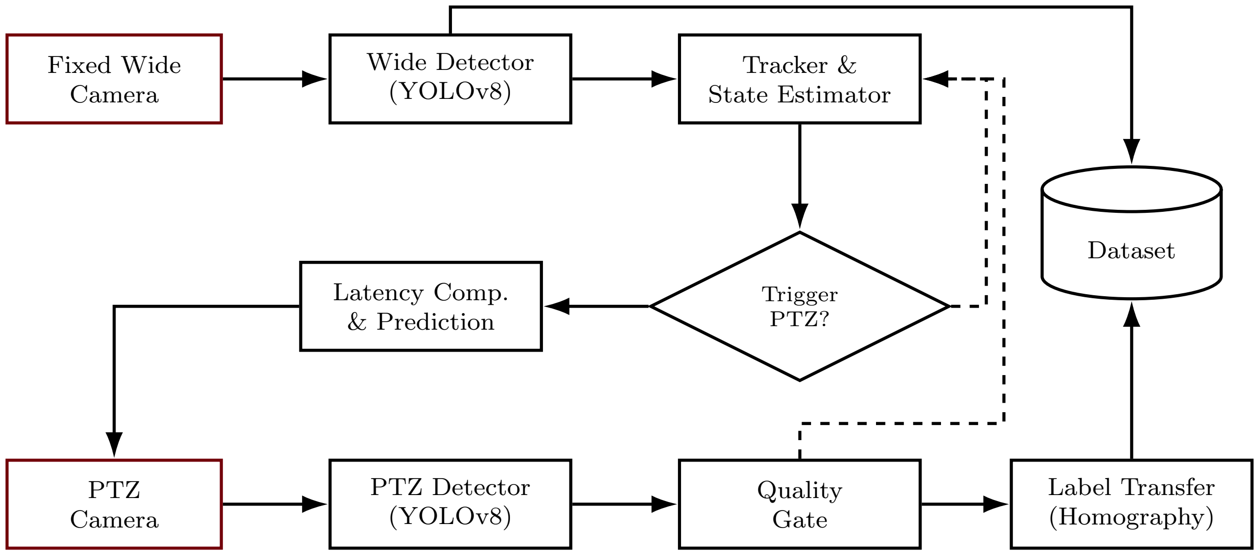

Fig. 6 Automated label-generation pipeline. Wide-field tracking selects candidates, PTZ verifies optically, and homography maps labels to the wide view.

Table 1 System configuration and measured detector throughput. All models use 640×640 **input with FP16 precision.**

Hardware & Software		*Detector Throughput*		
Component	Specification		Stationary (GPU)	PTZ (DLA)
Platform	Jetson Orin AGX	Model	YOLOv11s	YOLOv11n
Software	JetPack 4.7, CUDA 12.6	Inference (ms)	27.9	46.5
Cameras	2× FLIR M300C	Throughput (FPS)	35.9	21.5
Wide FOV	60°H × 45°V	Detection rate	100%	99%
Video codec	MJPEG	Avg confidence	0.553	0.529

become unambiguous at higher resolution, and the PTZ detector must independently confirm the target class above $\tau_{\text{confirm}} = 0.35$ for the sample to be accepted. Samples are also rejected when the PTZ detector class disagrees with the wide detector class. Additional quality gates for motion blur (Laplacian variance thresholding), exposure saturation, and frame-boundary rejection are defined in the pipeline architecture but are not yet active in the current deployment.

C. Dataset Schema

Each committed example stores the wide frame (or short clip), the transferred bounding box label, PTZ telemetry at capture time, both detector scores, and quality metrics (blur, saturation, occlusion flags).

VII. Experimental Validation

A. System Configuration

The complete system runs on an NVIDIA Jetson Orin AGX with JetPack 4.7 and CUDA 12.6. Table 1 summarizes the hardware and software configuration. The stationary wide camera provides a $60^\circ \times 45^\circ$ field of view for persistent surveillance, while the PTZ camera operates at configurable zoom levels of 25%, 35%, and 45% during tracking.

B. Detector Throughput

The two detector instances are split across compute units to avoid GPU time-slicing overhead. The stationary camera runs YOLOv11s on the GPU, where larger model capacity aids feature extraction on distant, low-resolution targets. The PTZ camera runs YOLOv11n on the DLA, where zoomed-in targets are easier to classify. This split achieves true parallel inference with no resource contention.

The GPU delivers 2.2–2.8× higher throughput than the DLA per model. In ablation benchmarks on YOLOv8n at 640×640, the GPU alone achieves 23.9 FPS (41.9 ms latency) while the DLA achieves 16.0 FPS (62.4 ms latency). Running both models on the GPU forces CUDA time-slicing, where context switches between the two inference streams degrade aggregate throughput below the sum of individual rates. The GPU/DLA split eliminates this contention because the DLA is a dedicated hardware accelerator with its own execution pipeline, enabling true parallel inference at 35.9 + 21.5 = 57.4 aggregate FPS (Table 1). The stationary detector achieves 100% detection rate with an average confidence of 0.553, while the PTZ detector achieves 99% detection rate with an average confidence of 0.529.

C. Tracking Controller

The PTZ centering loop uses a PID controller running at 30 Hz. Table 2 lists the controller parameters. PID gains were iteratively hand-tuned starting from K_p=3.0, K_i=0.1, K_d=0.5; K_p was halved to reduce overshoot, K_i reduced to prevent integral windup, and K_d doubled to increase damping. The controller applies inverse-square speed scaling with zoom level: at maximum zoom, PTZ speed is limited to 15% of the base rate, matching the narrower field of view to prevent the target from exiting the frame during corrections. A 2% centering deadzone suppresses jitter when the target is near frame center. Confidence thresholds are set asymmetrically: the acquisition threshold (0.40) is higher than the tracking threshold (0.35) to prevent the system from acquiring weak detections while allowing it to maintain lock through brief confidence dips. The 3.0 s lost threshold was selected to exceed two full reacquisition cycles at the 30 Hz control rate before reverting to Phase 1.

The tracker operates as a five-state machine: IDLE (home position, awaiting target), ACQUIRING (open-loop slew from stationary camera cue), TRACKING (closed-loop PID centering), REACQUIRING (progressive zoom-out search after brief loss), and LOST (return to home). The ACQUIRING→TRACKING transition fires on the first PTZ detection or after a 3.0 s coarse-tracking timeout, whichever occurs first. If the PTZ loses detection for 15 consecutive frames during TRACKING, the system enters REACQUIRING and steps through progressively wider zoom levels at 1.5 s intervals. A 5.0 s reacquisition timeout triggers the LOST state, which resets the PTZ to home and returns to IDLE.

Table 2 Tracking controller parameters.

Parameter	Value	Unit	Parameter	Value	Unit
PID gain K_p	1.5	—	Center deadzone	2	%
PID gain K_i	0.05	—	Position tolerance	0.2	deg
PID gain K_d	1.0	—	Lost threshold	3.0	s
Control rate	30	Hz	Acquire timeout	10	s
Acquire confidence	0.40	—	Reacquire timeout	5	s
Track confidence	0.35	—	Max PTZ speed	30	deg/s
Zoom levels	25, 35, 45	%	Min zoom scale	15	%

D. Video Codec Impact on Latency

Video codec selection has a direct impact on tracking latency. The system initially used H.264, which requires dependency frames (I/P/B) and introduces 2–4 frames of decode buffering latency (66–133 ms at 30 fps) before the decoder can deliver a complete image to the detector. Switching to MJPEG, which encodes each frame independently, eliminates this pipeline stall and reduces decode latency to under 5 ms per frame. This 60–130 ms reduction is significant relative to the total control loop period of 33 ms and directly improves closed-loop tracking responsiveness.

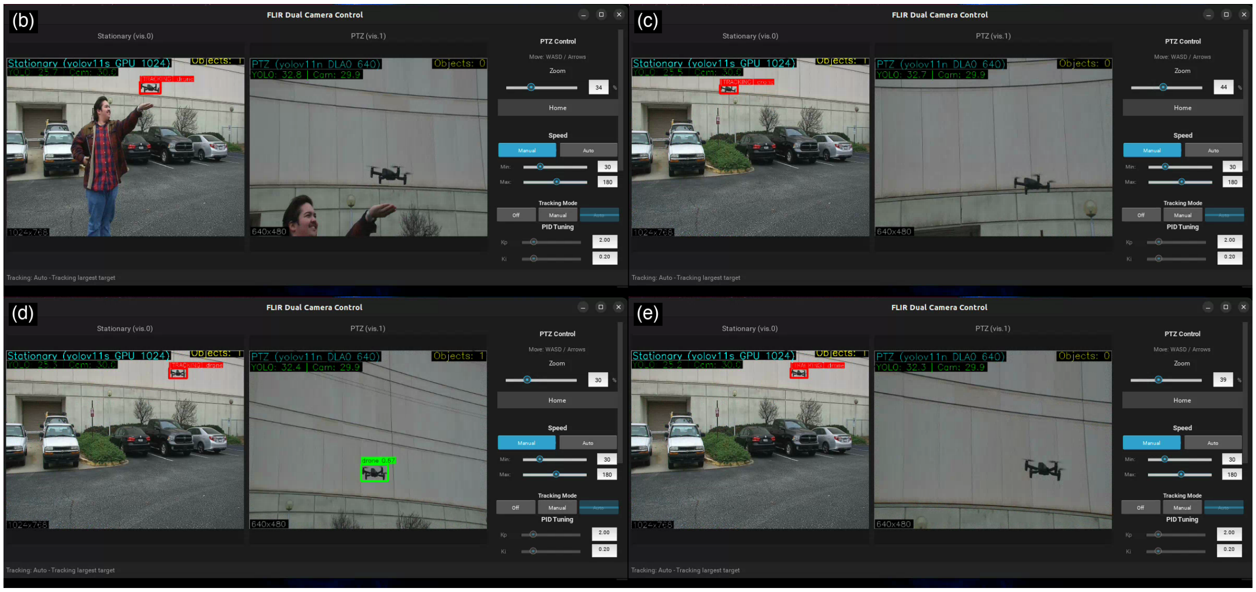

Fig. 7 Temporal montage from the 42-second drone tracking sequence. (a) Early tracking with PTZ slewing to target. (b) Active tracking at medium zoom with drone visible in both cameras. (c) PTZ detection at higher zoom with green bounding box on target. (d) Sustained tracking in final flight segment.

E. Drone Tracking Demonstration

To validate end-to-end system operation, we conducted an outdoor drone tracking test. The system autonomously acquired and tracked a small commercial drone over a 42-second flight, producing approximately 42,000 frames across both cameras. Figure 7 presents a temporal montage of selected frames spanning the tracking sequence.

The stationary camera detected the drone at wide-angle resolution, providing bearing cues to the PTZ controller. Upon first detection above the acquisition threshold, the system transitioned from IDLE to ACQUIRING: the PTZ slewed to the predicted bearing at 25% zoom (the widest setting, maximizing the probability of initial capture). Once the PTZ detector confirmed the target, the system entered the TRACKING state and the PID centering loop took control. If the PTZ lost detection for more than 15 consecutive frames (~0.75 s at 20 FPS), the system entered a REACQUIRING state, progressively zooming out through discrete levels (25%, 35%, 50%, 70%) at 1.5 s intervals to widen the search area. If reacquisition failed within 5.0 s, the PTZ returned to its home position and the system reverted to IDLE, awaiting the next stationary-camera cue.

During the 42-second flight, the system maintained continuous track without entering the REACQUIRING state, indicating stable PID lock throughout. The on-screen overlay (visible in Fig. 7) displays the model type, accelerator assignment, and inference resolution for each camera, confirming simultaneous GPU and DLA operation.

In the wide-angle view, the drone occupies approximately 20×15 pixels—insufficient for reliable classification. The PTZ view at 35–45% zoom resolves the target at approximately 120×90 pixels, a 6× increase in linear resolution that moves the target into the regime where YOLOv8 achieves 99.1% mAP50 on flying objects [6]. The stationary detector maintained continuous detection throughout the flight (100% detection rate, mean confidence 0.553), while the PTZ detector confirmed the target in 99% of zoomed frames (mean confidence 0.529).

F. In-Lab Airplane Detection

In-laboratory testing (Fig. 8) confirmed that the default YOLOv8n COCO "airplane" class detects large, high-resolution aircraft imagery displayed on a monitor but fails on small pixel footprints typical of the wide-angle surveillance view. This motivated training custom detection models on a drone detection dataset and on the Fine-Grained Visual Classification (FGVC) aircraft dataset, which provides ground-up RGB imagery of aircraft at a range of scales and orientations representative of the surveillance scenario.

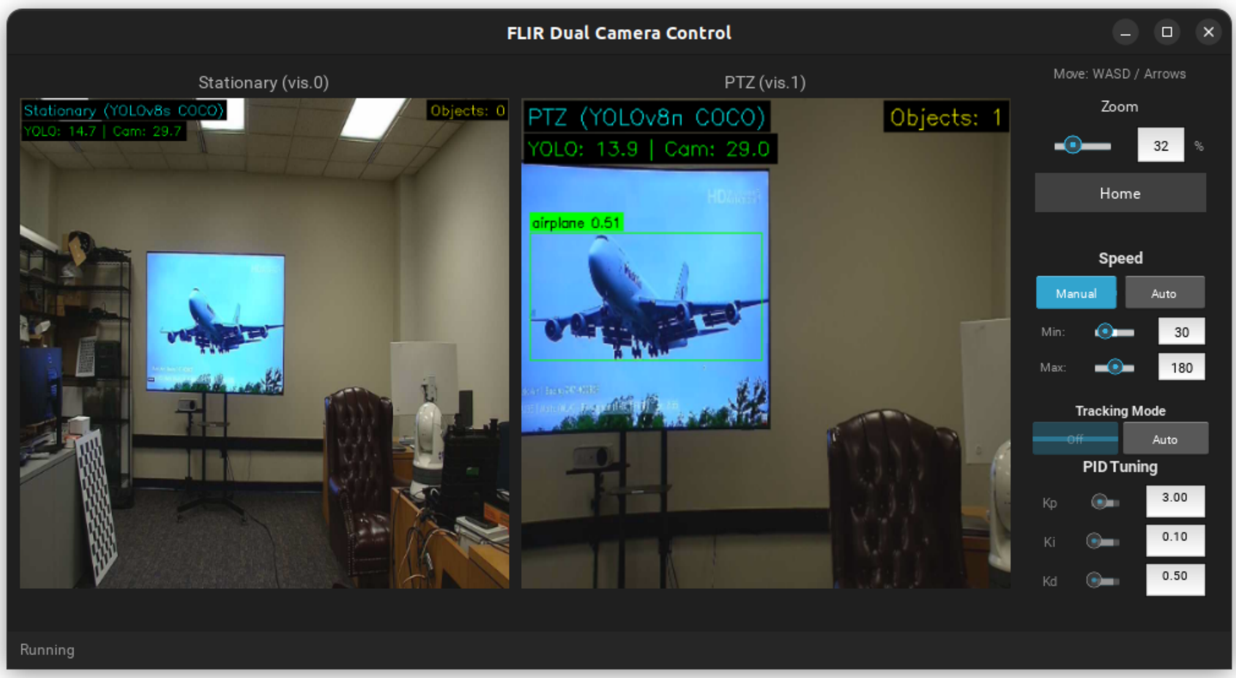

Fig. 8 In-laboratory validation. The PTZ camera detects and tracks an aircraft displayed on a monitor using YOLOv8n with COCO weights, confirming the detection pipeline and GUI control interface.

G. Airplanes Test Case

For the airplane surveillance application, the primary class is *airplane*. Hard negatives arise from birds, insects near the lens, distant drones, clouds with sharp edges, sensor noise, and specular highlights. The PTZ confirmation step naturally collects informative negatives: wide proposals that are rejected by PTZ as non-airplane are stored as hard negatives with contextual metadata.

Custom YOLO models trained on FGVC aircraft data are intended to replace the COCO-pretrained weights for the wide camera, targeting the small pixel sizes ($< 30 \times 30$) that COCO-trained models fail to detect. Benchmarking these custom models against default YOLO classes on a test set containing both long-distance and close-distance aircraft remains ongoing work.

VIII. Conclusion

This paper presented a dual-camera active acquisition system for automated small-object dataset construction. A fixed wide-angle camera with real-time detection provides persistent coverage, while a PTZ camera delivers optical zoom for verification. The system runs on an NVIDIA Jetson Orin AGX, achieving 35.9 FPS (GPU) and 21.5 FPS (DLA) through a parallel GPU/DLA inference split that eliminates resource contention. PID-controlled PTZ tracking with MJPEG codec selection minimizes pipeline latency for closed-loop operation. We demonstrated end-to-end functionality through a 42-second outdoor drone tracking test, where the system autonomously acquired and maintained lock on a small aerial target, resolving it from ~20 pixels in the wide view to ~120 pixels under optical zoom. The active acquisition paradigm provides optically verified imagery that neither digital zoom nor super-resolution can match, enabling automated dataset construction for small-object regimes where manual annotation is impractical.

Current limitations include single-PTZ availability (no simultaneous multi-target zoom), potential target loss during slew for fast-moving objects, and the small-parallax assumption for label transfer. Future work will evaluate label quality against manual annotation baselines, measure downstream mAP improvements, and explore multi-PTZ scheduling for higher-throughput acquisition.

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